Hanlei Zeng

XR / HCI Researcher • Interaction Design for Digital Cultural Heritage • Embodied Interaction & Spatial Experience Research

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Research Interests

My research focuses on **embodied interaction design at the intersection of HCI, XR, and digital cultural heritage**, exploring how **vision-based markerless gesture interaction, spatial computing, and natural user interfaces** can enhance interaction between people and digital environments. Through a **Research through Design** methodology, I develop interactive prototypes and study their applications in **digital heritage exhibitions, immersive experiences, and spatial narrative design**. I am also interested in the potential of **interactive lighting as a digital medium**, exploring mappings between light, bodily movement, and cultural content.

Research Experience

Gesture-Driven Embodied Interaction for Digital Heritage

Independent research & prototyping | 2025 – Present

Exploring how vision-based markerless gesture interaction can foster user participation and cultural experience, centered on **HCI, XR, and digital heritage interaction**.

- Developed browser-based 3D heritage interaction prototypes (**JavaScript, MediaPipe, Three.js**) around bronze artifacts and Dunhuang digital heritage, combining 3D models, gesture input, and visual feedback to explore embodied cultural participation.
- Developed mixed-reality spatial interaction prototypes (**Unity, C#, MRTK, HoloLens 2**), building user-needs models with grounded theory to realize spatial visualization and interactive customization. The project originated from the Zhenjiang Municipal Key R&D Program (GY2021014).
- Principal investigator of **JAT251371 (VR-based collaborative lighting design)**, exploring the combination of XR spatial interaction and interactive lighting design.

Research Methods

Qualitative: Grounded theory, semi-structured interviews, participatory design, contextual inquiry
Quantitative: Experimental design, SEM, DEMATEL, questionnaire analysis, usability evaluation
Design research: Research through Design, interactive prototype development, iterative design

Research Projects

- **PI**: JAT251371, VR-based collaborative lighting design.
- **Participant**: GY2021014, R&D of a mixed-reality-based personalized customization system for modified vehicles (Zhenjiang Municipal Key R&D Program).
- **Participant**: 2020 Industrial Center Student Innovation Practice Fund project, cross-regional medical system design based on 5G networks and virtual reality.

Research Output

Papers

Sha, C., Fang, Y., **Zeng, H.**, Han, R., & Donkor, E. K. N. (2026). A mixed reality-based personalization system for recreational vehicle interior customization. **Tehnički vjesnik – Technical Gazette.**

Sha, C., **Zeng, H.**, Wu, B., et al. (2022). Research on the application of mixed reality technology in the field of art design. **SHS Web of Conferences, 145.**

Zeng, H. (2023). Design research on a mixed reality-based customization system for RV interiors. Master’ s Thesis, Jiangsu University.

Patents & Software Copyright

Granted invention patent: Mixed-reality-based RV interior customization method and system (CN114782660A).

Software copyright: MR RV Interior Personalization System (2022SR0889245).

Work Experience

1. **Quanzhou Huaguang College**, Director • Digital Intangible Cultural Heritage Design and Production | 2025.11 – Present
   Lead teaching and curriculum research in digital intangible cultural heritage design and production; teach 3D character design, digital sculpting and painting, graphic drawing fundamentals, and digital creative product design; supervise student 3D content production and digital visual work.
2. **OSRAM / LEDVANCE**, Hardware Product Manager | 2024.10 – 2025.10
   Led development and launch of 9 lighting product series (nearly 100 SKUs), generating approximately US$8 million in cumulative sales; contributed to international projects such as Zayed Airport, applying light-distribution analysis, environmental simulation, and spatial experience design.
3. **LEEDARSON**, Hardware Product Manager | 2023.07 – 2024.07
   Conducted user and market research for North American and European product lines using Amazon sales data, word-frequency analysis, and Discord user communities; applied DEMATEL to extract user requirements; managed development from concept to mass production, launching scenario-based lighting products with more than US$30 million in cumulative sales.
4. **TCL**, Smart Home Product Manager Intern | 2022.05 – 2022.12
   Conducted user interviews and scenario research on sleep, home-entertainment, and air-purification use cases; produced more than 50 interaction prototypes and user flows; explored multi-device coordination in smart-home environments.

Education

Jiangsu University | M.F.A. in Digital Art Design | 2020 – 2023
National postgraduate entrance examination: 402/500; ranked first in both the written examination and the interview.

Hunan University of Technology Science and Technology College | B.A. in Product Design | 2016 – 2020

Technical Skills

XR / Interaction: Unity • C# • HoloLens 2 • MRTK • JavaScript • MediaPipe • Three.js
3D / Digital Media: ZBrush • Maya • Rhino • Cinema 4D • Blender • KeyShot
Design Tools: Figma • Photoshop • Illustrator
Research Methods: Grounded Theory • SEM • DEMATEL • User Study • Usability Evaluation
Lighting & AI Workflow: DIALux • Generative AI • Python Workflow Automation

Awards

Bauhaus International Design Competition (Silver) • Milan Design Week (Second Prize) • Rural Museum Narrative Design (First Prize) • Guoqing Cup (Second Prize) • Cross-Strait Emerging Design Award, Huacan (Third Prize) • National College Digital Art Design Competition (Third Prize) • Jinshan Cultural & Creative Design Competition (Excellence Award) • Chinese Ethnic Visual Identity Design Competition (Third Prize)